Martin Ward

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Address1: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html

GET/REPLY

117	3.586526	192.168.2.22	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
119	3.622320	128.119.245.12	192.168.2.22	HTTP	540	HTTP/1.1 200 OK (text/html)

```
> Frame 117: 526 bytes on wire (4208 bits), 526 bytes captured (4208 bits) on interface \Device\NPF_{41D04EE8-9D5C-4E0B-AE1A-ED96B40
> Ethernet II, Src: Intel_a4:01:0a (f0:d4:15:a4:01:0a), Dst: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)
> Internet Protocol Version 4, Src: 192.168.2.22, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 59193, Dst Port: 80, Seq: 1, Ack: 1, Len: 472
> Hypertext Transfer Protocol
```

GET - Six Addresses

Source IP: 192.168.2.22

Destination IP: 128.119.245.12

Source Port: 59193

Destination Port: 80 (HTTP)

Source: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Destination: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

REPLY - Six Addresses

Source IP: 128.119.245.12

Destination IP: 192.168.2.22

Source Port: 80 (HTTP)

Destination Port: 59193

Source: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

Destination: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Time:

Time	Source	Destination	Protocol	Lengt	Info
2024-09-28 03:01:34.403589	192.168.2.22	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
2024-09-28 03:01:34.439383	128.119.245.12	192.168.2.22	HTTP	540	HTTP/1.1 200 OK (text/html)

34.439383 - 34.439383 = 0.035794 seconds

First URL:

SYN, SYN-ACK, ACK

tcp.port==59193

No.	Time	Source	Destination	Protocol	Lengt	Info
111	3.550582	192.168.2.22	128.119.245.12	TCP	66	59193 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
113	3.586336	128.119.245.12	192.168.2.22	TCP	66	80 → 59193 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
115	3.586393	192.168.2.22	128.119.245.12	TCP	54	59193 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0

FIN/RST

204	8.627421	128.119.245.12	192.168.2.22	TCP	54	80 → 59193 [FIN, ACK] Seq=487 Ack=473 Win=30336 Len=0
205	8.627467	192.168.2.22	128.119.245.12	TCP	54	59193 → 80 [ACK] Seq=473 Ack=488 Win=130816 Len=0

Address2: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html

GET/REPLY

78	2024-09-28 0…	192.168.2.22	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
82	2024-09-28 0…	128.119.245.12	192.168.2.22	HTTP	784	HTTP/1.1 200 OK (text/html)

GET - Six Addresses

Source IP: 192.168.2.22

Destination IP: 128.119.245.12

Source Port: 59777

Destination Port: 80 (HTTP)

Source: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Destination: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

REPLY - Six Addresses

Source IP: 128.119.245.12

Destination IP: 192.168.2.22

Source Port: 80 (HTTP)

Destination Port: 59777

Source: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

Destination: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Time:

78	2024-09-28 07:22:41.624666	192.168.2.22	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
82	2024-09-28 07:22:41.656265	128.119.245.12	192.168.2.22	HTTP	784	HTTP/1.1 200 OK (text/html)

41.656265 - 41.624666 = 0.031599 seconds

Second URL:

SYN, SYN-ACK, ACK

75	2024-09-28 07:22:41.594101	192.168.2.22	128.119.245.12	TCP	66	59777 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
76	2024-09-28 07:22:41.624228	128.119.245.12	192.168.2.22	TCP	66	80 → 59777 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=128
77	2024-09-28 07:22:41.624310	192.168.2.22	128.119.245.12	TCP	54	59777 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0

FIN/RST

404 not found, and client machine send ACK response, otherwise response should be been similar to address 1 & 3.

75	2024-09-28 07:22:41.594101	192.168.2.22	128.119.245.12	TCP	66	59777 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
76	2024-09-28 07:22:41.624228	128.119.245.12	192.168.2.22	TCP	66	80 → 59777 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1
77	2024-09-28 07:22:41.624310	192.168.2.22	128.119.245.12	TCP	54	59777 → 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
78	2024-09-28 07:22:41.624666	192.168.2.22	128.119.245.12	HTTP	526	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
81	2024-09-28 07:22:41.655281	128.119.245.12	192.168.2.22	TCP	54	80 → 59777 [ACK] Seq=1 Ack=473 Win=30336 Len=0
82	2024-09-28 07:22:41.656265	128.119.245.12	192.168.2.22	HTTP	784	HTTP/1.1 200 OK (text/html)
83	2024-09-28 07:22:41.684463	192.168.2.22	128.119.245.12	HTTP	472	GET /favicon.ico HTTP/1.1
84	2024-09-28 07:22:41.716288	128.119.245.12	192.168.2.22	HTTP	538	HTTP/1.1 404 Not Found (text/html)
92	2024-09-28 07:22:41.770893	192.168.2.22	128.119.245.12	TCP	54	59777 → 80 [ACK] Seq=891 Ack=1215 Win=130048 Len=0

Address3: http://info.cern.ch/

GET/REPLY

110	2024-09-28 07:34:45.450320	192.168.2.22	188.184.67.127	HTTP	481	GET / HTTP/1.1
115	2024-09-28 07:34:45.559309	188.184.67.127	192.168.2.22	HTTP	932	HTTP/1.1 200 OK (text/html)
119	2024-09-28 07:34:45.593346	192.168.2.22	188.184.67.127	HTTP	422	GET /favicon.ico HTTP/1.1
123	2024-09-28 07:34:45.702453	188.184.67.127	192.168.2.22	HTTP	428	HTTP/1.1 200 OK (image/vnd.microsoft.icon)

GET - Six Addresses

Source IP: 192.168.2.22

Destination IP: 188.184.67.127

Source Port: 59924

Destination Port: 80 (HTTP)

Source: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Destination: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

REPLY - Six Addresses

Source IP: 188.184.67.127

Destination IP: 192.168.2.22

Source Port: 80 (HTTP)

Destination Port: 59924

Source: SagemcomBroa_87:60:4e (0c:ac:8a:87:60:4e)

Destination: Intel_a4:01:0a (f0:d4:15:a4:01:0a)

Time:

110	2024-09-28 07:34:45.450320	192.168.2.22	188.184.67.127	HTTP	481	GET / HTTP/1.1
115	2024-09-28 07:34:45.559309	188.184.67.127	192.168.2.22	HTTP	932	HTTP/1.1 200 OK (text/html)

45.559309 - 45.450320 = 0.108989 seconds

Third URL:

SYN, SYN-ACK, ACK

102	2024-09-28 07:34:45.230403	192.168.2.22	188.184.67.127	TCP	66	59924 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
104	2024-09-28 07:34:45.343305	188.184.67.127	192.168.2.22	TCP	58	80 → 59924 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0 MSS=1434
106	2024-09-28 07:34:45.343568	192.168.2.22	188.184.67.127	TCP	54	59924 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0

FIN/RST

118	2024-09-28 07:34:45.559783	192.168.2.22	188.184.67.127	TCP	54	59924 → 80 [FIN, ACK] Seq=428 Ack=880 Win=63362 Len=0
120	2024-09-28 07:34:45.667442	188.184.67.127	192.168.2.22	TCP	54	80 → 59924 [ACK] Seq=880 Ack=429 Win=31693 Len=0

ARP:

Address Resolution Protocol (ARP) is a network protocol used to map an IP address to a MAC (Media Access Control) address.

The ARP request i send to ask for a MAC address associated with that IP.

ARP protocol goes in pairs of ARP Request to ARP Reply for example ARP requests send a boardcast who has X ip send to Y, whoever has the X ip will then forward its max address to Y.

Its a kind of addres mapping with the results being cached at times to avoid repeat requests.